

Table OM1

Chalcophile elements (PGE, Cu, and Ni) and S contents of whole-rock samples, and Cu isotopic compositions and Cu/Fe ratios of chalcopyrite separates from the Jinchuan deposit, NW China.

Sample no.	Locations	Line No	Elevation (m)	Mineralization types	$\delta^{65}\text{Cu}$ (‰)	2SD	Cu/Fe	Cu (ppm)	Ni (ppm)	S (wt. %)	Pt (ppm)	Pd (ppm)	Rh (ppm)	Ru (ppm)	Ir (ppm)	Os (ppm)
2j5-1	Segment II	5	1040	Net-textured	0.02	0.05	0.91	35260	26330	11.0	60.5	341.0	11.5	11.4	20.8	28.0
2j5-2	Segment II	5	1040	Net-textured	-0.03	0.07	0.90	3596	29203	9.9	4450.0	137.0	25.1	25.5	39.3	66.3
2j4-4	Segment II	4	1040	Net-textured	0.60	0.05	0.94	5363	31970	9.7	16.4	372.0	14.7	11.3	21.8	26.3
2j4-3	Segment II	4	1040	Disseminated	0.29	0.05	0.93	4350	20780	3.5	10.3	243.5	11.2	10.5	16.5	12.3
2j4-2	Segment II	4	1040	Net-textured	0.57	0.09	0.98	27290	28567	9.4	121.0	802.0	1.5	0.6	1.4	2.5
2j3-7	Segment II	3	1040	Country rocks	0.23	0.05	nd	16	19	0.2	nd	nd	nd	nd	nd	nd
2j3-8	Segment II	3	1040	Country rocks	0.21	0.05	nd	27	16	0.2	nd	nd	nd	nd	nd	nd
2j3-10	Segment II	3	1040	Country rocks	0.22	0.05	nd	22	14	0.2	nd	nd	nd	nd	nd	nd
2j3-11	Segment II	3	1040	Country rocks	0.23	0.05	nd	25	15	0.3	nd	nd	nd	nd	nd	nd
2J1-1	Segment II	1	1040	Net-textured	0.39	0.09	0.96	15600	46131	9.7	15.4	486.0	20.4	24.0	34.4	46.7
2J1-2	Segment II	1	1040	Net-textured	0.57	0.05	0.90	43610	42411	10.3	54.7	320.0	15.7	9.9	17.5	22.6
2J1-4	Segment II	1	1040	Disseminated	0.69	0.05	0.93	6439	21825	5.9	367.0	131.0	2.4	0.9	2.0	3.6
2j1-1	Segment II	1.25	1040	Net-textured	-0.21	0.08	0.92	70980	14056	11.1	342.0	337.0	10.8	8.2	18.1	12.9
2j1-2	Segment II	1.25	1040	Net-textured	-0.47	0.05	0.95	2044	17225	9.8	695.0	82.8	12.3	6.9	13.2	5.5
2j1-3	Segment II	1.25	1040	Net-textured	-0.27	0.10	0.92	70980	14056	9.6	29.3	367.0	1.2	1.0	1.1	2.6
2j1-4	Segment II	1.25	1040	Net-textured	0.15	0.05	0.90	72070	6777	9.2	555.0	371.0	0.7	0.7	1.0	1.5
j1-1	Segment I	1	1040	Net-textured	0.00	0.06	0.95	50980	21189	9.9	54.5	151.0	1.4	2.5	2.6	5.1
j1-2	Segment I	1	1040	Net-textured	0.62	0.05	0.92	21990	21741	7.7	269.0	505.0	1.7	2.6	4.6	3.7
j3-1	Segment I	3	1040	Disseminated	0.32	0.05	0.93	4777	16854	6.0	260.0	205.0	12.9	17.8	33.1	23.8
j3-4	Segment I	3	1040	Disseminated	0.41	0.05	0.92	5432	12773	5.3	17.5	79.8	9.7	9.5	13.8	20.2
j3-6	Segment I	3	1040	Net-textured	0.19	0.05	0.94	3980	18200	6.6	477.0	265.0	9.2	9.7	15.0	10.3
j5-1	Segment I	5	1040	Disseminated	0.58	0.05	0.95	6125	10435	3.9	157.0	114.0	7.6	15.8	25.4	23.9

Table OM1 (continued)

Sample no.	Locations	Line No	Elevation (m)	Mineralization types	$\delta^{65}\text{Cu}$ (‰)	2SD	Cu/Fe	Cu (ppm)	Ni (ppm)	S (wt. %)	Pt (ppm)	Pd (ppm)	Rh (ppm)	Ru (ppm)	Ir (ppm)	Os (ppm)
j5-3	Segment I	5	1040	Disseminated	0.44	0.05	0.97	6869	15328	5.0	76.2	86.4	8.5	17.4	22.6	36.8
j5-4	Segment I	5	1040	Disseminated	0.57	0.05	0.91	7200	16700	5.0	36.5	28.5	5.3	13.5	32.2	24.2
j5-5	Segment I	5	1040	Disseminated	0.53	0.06	0.90	1050	6248	1.6	16.0	31.6	4.2	13.8	21.3	26.3
j5-6	Segment I	5	1040	Disseminated	0.38	0.05	0.89	2375	4914	1.2	32.5	43.6	3.0	6.9	7.8	12.8
j5-9	Segment I	5	1040	Net-textured	-0.47	0.05	0.91	47970	9533	7.4	265.0	158.0	13.8	24.3	54.0	101.0
j5-9 (replicate)	Segment I	5	1040	Net-textured	-0.44	0.05	0.93	nd	nd	nd	nd	nd	nd	nd	nd	nd
j5-11	Segment I	5	1040	Net-textured	0.65	0.05	0.98	7360	32213	7.0	23.9	156.0	7.9	8.2	12.1	25.6
j5-13	Segment I	5	1040	Net-textured	1.29	0.05	0.92	42840	17469	7.3	56.0	151.0	0.4	0.2	0.6	1.5
j5-13 (replicate)	Segment I	5	1040	Net-textured	1.18	0.08	0.92	nd	nd	nd	nd	nd	nd	nd	nd	nd
j5-14	Segment I	5	1040	Net-textured	0.56	0.05	0.96	42760	38902	9.3	24.9	345.0	11.3	4.9	10.6	13.3
j6-1	Segment I	6	1040	Disseminated	0.96	0.05	0.93	43500	13200	2.5	534.0	76.0	17.0	22.0	35.9	37.3
j6-2	Segment I	6	1040	Net-textured	0.15	0.05	0.90	16210	15510	10.6	769.0	88.1	25.7	35.6	46.9	34.9
j6-3	Segment I	6	1040	Net-textured	0.37	0.05	0.98	58800	9800	9.5	266.0	89.0	6.5	2.4	18.2	9.9
j6-5	Segment I	6	1040	Net-textured	-0.2	0.05	0.90	5615	32044	9.2	20.1	476.0	14.2	20.9	27.9	38.0
j6-5 (replicate)	Segment I	6	1040	Net-textured	-0.25	0.05	0.91	nd	nd	nd	nd	nd	nd	nd	nd	nd
j6-6	Segment I	6	1040	Net-textured	0.33	0.05	0.94	56670	17416	9.3	1219.0	453.0	2.1	2.6	3.1	4.2
j6-7	Segment I	6	1040	Net-textured	0.67	0.05	1.02	73300	5030	9.4	238.0	164.0	37.8	1.4	1.2	0.7
j6-9	Segment I	6	1040	Net-textured	1.05	0.13	1.08	57900	15434	8.6	302.0	231.0	1.2	1.6	2.4	3.8
j6-10	Segment I	6	1040	Net-textured	0.86	0.05	0.98	49510	10366	8.6	391.0	125.0	0.9	2.0	1.1	2.5
j6-11	Segment I	6	1040	Net-textured	0.51	0.05	0.99	62700	14522	9.4	1584.0	472.0	1.1	0.5	0.7	2.0
j7-1	Segment I	7	1040	Net-textured	0.32	0.06	0.91	70530	15412	10.7	244.0	267.0	3.1	5.4	7.2	4.3
j7-2	Segment I	7	1040	Net-textured	0.45	0.05	1.09	71500	10296	9.2	205.0	235.0	1.0	0.8	0.9	1.2
j7-4	Segment I	7	1040	Net-textured	0.43	0.06	1.01	53300	8909	7.3	295.0	298.0	2.7	5.3	5.3	4.2
j20-12-1	Segment I	20	1220	Disseminated	0.26	0.05	0.98	4480	3132	1.3	7.8	4.7	0.4	0.6	5.8	1.1

Table OM1 (continued)

Sample no.	Locations	Line No	Elevation (m)	Mineralization types	$\delta^{65}\text{Cu}$ (‰)		2SD	Cu/Cu/Fe	Cu (ppm)	Ni (ppm)	S (wt. %)	Pt (ppm)	Pd (ppm)	Rh (ppm)	Ru (ppm)	Ir (ppm)	Os (ppm)
j25-1	Segment I	25	1220	Disseminated	0.26	0.05	0.91	6717	10274	3.1	84.6	42.8	4.3	7.5	9.7	19.4	
j25-3	Segment I	25	1220	Disseminated	0.4	0.05	0.90	13100	12434	4.6	111.0	128.0	16.3	24.5	36.9	35.6	
j20-1	Segment I	20	1220	Massive	-0.36	0.05	1.13	20440	72684	25.1	93.2	623.0	117.0	339.0	369.0	643.0	
j20-2	Segment I	20	1220	Massive	0.09	0.05	1.00	7844	87620	30.1	67.7	197.0	135.0	377.0	419.0	357.0	
j20-3	Segment I	20	1220	Massive	-0.46	0.05	0.99	34000	71656	28.9	150.0	705.0	163.0	587.0	542.0	625.0	
j20-4	Segment I	20	1220	Massive	-0.49	0.12	0.99	94990	42560	20.1	146.0	53.0	35.2	149.0	143.0	220.0	
j20-5	Segment I	20	1220	Massive	-0.22	0.07	0.96	115900	34492	24.0	2049.0	412.0	50.9	463.0	402.0	551.0	
j20-5 (replicate)	Segment I	20	1220	Massive	-0.19	0.05	0.95	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd
j20-6	Segment I	20	1220	Massive	-0.44	0.05	0.97	50470	49629	21.8	1089.0	697.0	63.3	217.0	344.0	228.0	
j20-7	Segment I	20	1220	Massive	-0.83	0.05	0.97	38390	62943	26.8	38125.0	936.0	333.0	225.0	405.0	368.0	
j24-6	Segment I	24	1220	Massive	-0.91	0.05	0.90	4229	89570	34.4	119.0	365.5	154.0	337.0	192.0	405.0	
j24-8	Segment I	24	1220	Massive	-0.6	0.05	1.02	1485	77645	34.8	66.3	464.0	139.0	366.0	399.0	435.0	
j24-7	Segment I	24	1220	Massive	-0.38	0.05	0.90	1767	72589	34.8	74.5	399.0	110.0	273.0	260.0	367.0	

Notes: Copper isotope data are expressed in per mil relative to NIST 976. nd = not determined. Replicate = repeat sample dissolution, purified column chemistry and MC-ICPMS analysis.